Upload Heart Disease Dataset (CSV)

Upload

200MB per file • CSV

Data & Analysis Settings

Include 'Alcohol Consumption'

Excluding Alcohol Consumption retains ~9,900 records.

Use Medical Binning (Ordinal)

Show cleaned dataset

Heart Disease Risk Factor Analysis

Linear Correlation vs. Model-Based Feature Importance

Compare how variables rank under simple linear assumptions (Pearson) versus a non-linear machine learning model (Random Forest).

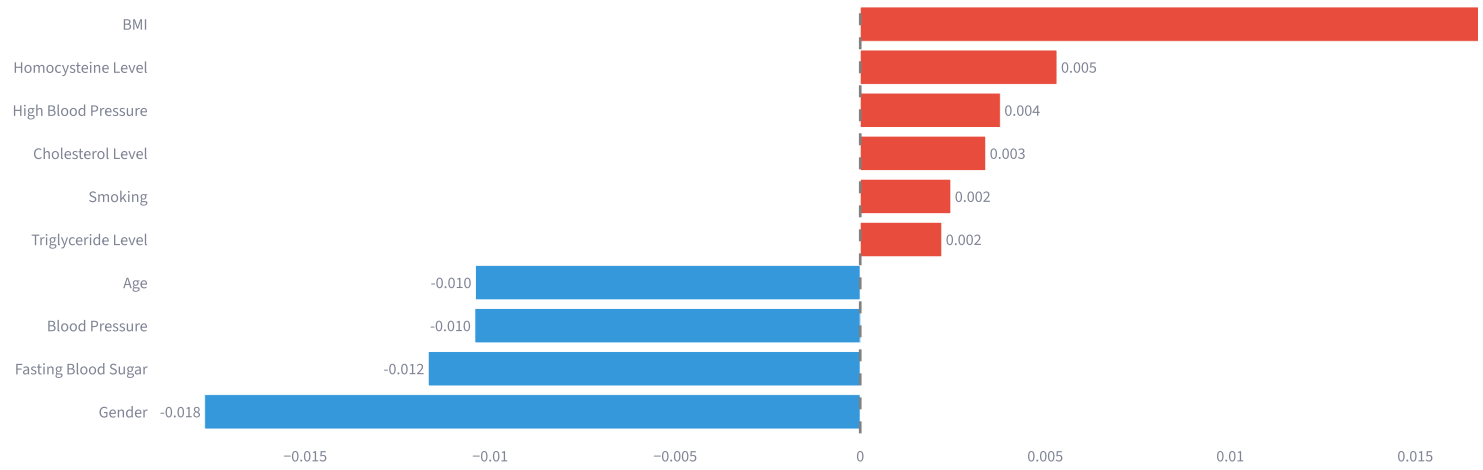
Medical Binning is ON: Continuous variables converted to ordinal risk levels (1-5).

Dataset Overview

Total Valid Records	Heart Disease Cases	No Heart Disease	Heart Disease Rate
9,762	1,954	7,808	20.02%

1. Pearson Correlation (Linear Association)

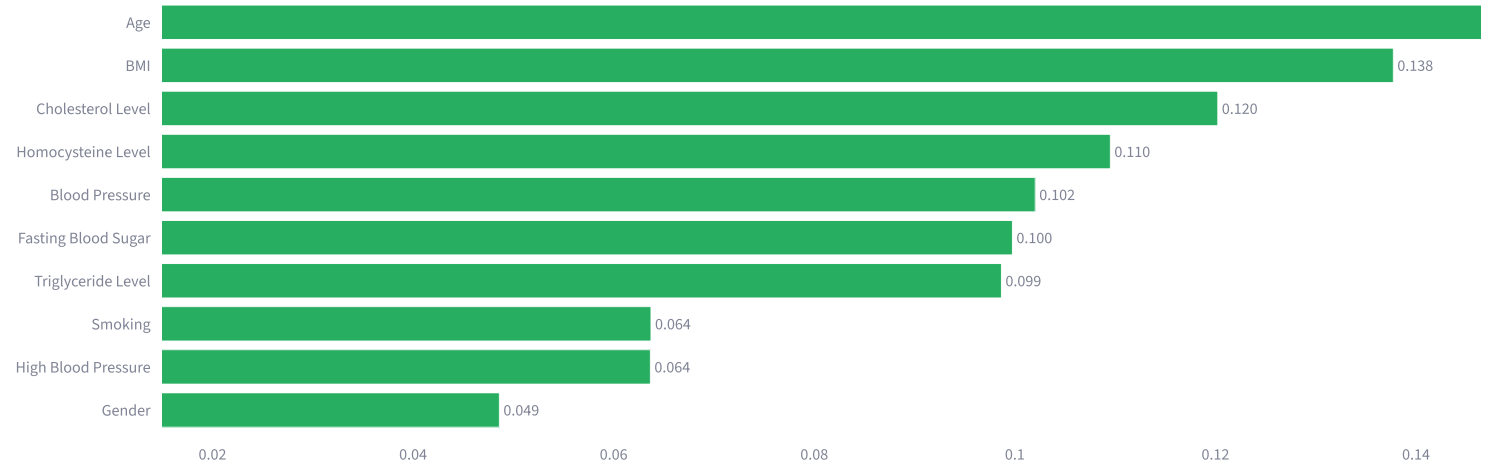
Pearson Correlation with Heart Disease



🌲 2. Random Forest Feature Importance (Non-Linear)

Why Random Forest Feature Importance? Pearson correlation only captures linear relationships. In medicine, risk factors often have complex, non-linear interactions. **Feature Importance** from a Random Forest model measures how much each variable contributes to the model's predictive accuracy. *(Note: The model is trained with* `class_weight='balanced'` *to account for class imbalance).*

Random Forest Feature Importance



📋 Combined Statistical Summary

Compare how variables rank differently under linear correlation vs. machine learning importance.

Variable	Pearson r	Abs Pearson	Feature Importance
Age	-0.0104	0.0104	
BMI	0.0223	0.0223	
Cholesterol Level	0.0034	0.0034	
Homocysteine Level	0.0053	0.0053	
Blood Pressure	-0.0104	0.0104	
Fasting Blood Sugar	-0.0117	0.0117	
Triglyceride Level	0.0022	0.0022	
Smoking	0.0024	0.0024	
High Blood Pressure	0.0038	0.0038	
Gender	-0.0177	0.0177	

Download Combined Results

Processed Dataset

BMI	Blood Pressure	Homocysteine Level	Triglyceride Level	Cholesterol Level	Gender	Smoking	Age	High Blood Pressure	Fasting Blood Su
3	5	4	2	4	0	0	4	0	
3	3	3	4	3	1	0	3	0	
2	3	2	4	4	0	1	2	1	
2	5	2	4	4	1	1	3	1	
3	5	3	2	4	1	1	1	0	
1	3	4	2	2	0	1	5	0	
4	5	4	4	2	0	1	2	0	
4	4	2	4	4	0	0	3	1	
4	5	2	3	4	1	1	4	0	
4	4	4	2	4	0	0	2	0	

Note: Feature importance indicates how useful a variable was to the model, not a causal relationship. Alcohol Consumption has a high missing rate; toggling it on/off demonstrates the trade-off between variable coverage and dataset size.